Earth

Earth is the third planet from the Sun and the only astronomical object known to harbor life. This is made possible by Earth being an ocean world, the only one in the Solar System sustaining liquid surface water. Almost all of Earth's water is contained in its ocean, which covers 70.8% of Earth's crust. The remaining 29.2% of Earth's crust is land, which is predominantly located within Earth's land hemisphere in the form of continental landmasses. Most of Earth's land is at least somewhat humid and covered by vegetation, while large ice sheets at Earth's polar deserts retain more water than Earth's groundwater, lakes, rivers, and atmospheric water combined. Earth's crust consists of slowly moving tectonic plates, which interact to produce mountain ranges, volcanoes, and earthquakes. Earth has a liquid outer core that generates a magnetosphere capable of deflecting most of the destructive solar winds and cosmic radiation.

Earth has a dynamic atmosphere, which sustains Earth's surface conditions and protects it from most meteoroids and ultraviolet light at entry. It is composed primarily of nitrogen and oxygen. Water vapor is widely present in the atmosphere, forming clouds that cover most of the planet. The water vapor acts as a greenhouse gas and, together with other greenhouse gases in the atmosphere, particularly carbon dioxide (CO_2), creates the conditions for both liquid surface water and water vapor to persist via the capturing of energy from the Sun's light. This process maintains the current average surface temperature of 14.76 °C (58.57 °F), at which water is liquid under normal atmospheric pressure. Differences in the amount of captured energy between geographic regions (as with the equatorial region receiving more sunlight than the polar regions) drive atmospheric and ocean currents, producing a global climate system with

Earth

Earth as seen from Meteosat-12 in 2025, during the March equinox

Designations	
Alternative names	The world · The globe · Terra · Tellus · Gaia
Adjectives	Earthly · Terrestrial · Terran · Tellurian
Symbol	⊕ and ♁[1]
Orbital characteristics	
	Epoch J2000[n 1]
Aphelion	152 097 597 km
Perihelion	147 098 450 km[n 2]
Semi-major axis	149 598 023 km[2]
Eccentricity	0.016 7086[2]
Orbital period (sidereal)	365.256 363 004 d[3] (1.000 017 420 96 a_j)
Average orbital speed	29.7827 km/s[4]
Mean anomaly	358.617°
Inclination	7.155° – Sun's equator; 1.578 69° – invariable plane;[5] 0.000 05° – J2000 ecliptic
Longitude of ascending node	−11.260 64° – J2000 ecliptic[4]